

Table 3: Learning objectives (Theory) of Course

Paper 1 Fundamental Dravyaguna									
A3 Course outcome	B3 Learning Objective (At the end of the session, the students should be able to)	C3 Domain/sub	D3 Must to know / desirable to know / Nice to know	E3 Level Does/ Shows how/ Knows how/ Know	F3 T-L method	G3 Assessment (Refer abbreviations)	H3 Formative/summative	I3 Term	J3 Integration
Topic 1 1.Dravyaguna Vigyana (Lecture :1 hours, Non lecture: 1 hours)									
CO1,CO3	Define Dravyaguna Vigyana.	CK	MK	K	L&PP T	P-VIVA	F&S	I	
CO1,CO3	Explain the role of Dravyaguna Vigyana in clinical practice.	CC	MK	KH	L&G D	P-VIVA,C-INT	F&S	I	
Topic 2 2.Dravya (Lecture :5 hours, Non lecture: 4 hours)									
CO1,CO5,CO8	Classify Dravya based on Utpatti, Yonibheda, Prayogabheda & Prabhavbheda.	CK	MK	K	L&PP T,LS	T-OBT,P-VIVA,P-POS	F&S	I	
CO1,CO5,CO8	Classify Dravya based on Doshagnabheda, Rasabheda, Karmbheda (Dashemani & Sushrutokta Gana)	CK	MK	K	L&PP T,LS	T-OBT,P-VIVA,P-POS	F&S	I	
CO1,CO3	Describe Panchabhoutikatwa of Dravya.	CC	MK	KH	DIS,P ER	T- EW,T-OBT,P-VIVA	F&S	I	
Topic 3 3. Guna (Lecture :4 hours, Non lecture: 2 hours)									

CO1,CO2,CO3	Define Guna and describe its Panchabhoutikatva.	CK	MK	K	L&PP T,DIS ,LS	T- EW,P-VIV A,PRN,QZ	F&S	I	
CO1	State characteristics of Gurvadi Guna.	CK	MK	K	L&PP T,SD L	P-VIVA,PRN, QZ	F&S	I	
CO1,CO2,CO3	Discuss Gurvadiguna in context to its Karma on Dosha, Dhatu and Mala with examples.	CC	MK	KH	L&PP T,L& GD,F C	T- EW,P-VIV A,PRN,OSPE	F&S	I	
CO1,CO2,CO3	Enumerate & Summarize, clinical application of Gurvadi Guna with research updates.	CC	MK	KH	L&G D,BL, LS	T- EW,P- VIVA,PRN	F&S	I	
CO1,CO2,CO3	Enlist Paradiguna with examples.	CK	MK	K	L&G D,PE R	T-EMI,P-VIV A,PRN,P-POS	F&S	I	
CO1,CO2,CO3	Interpret clinical applications of Paradiguna.	CAP	MK	KH	L&PP T,DIS	T-EMI,P-VIV A,PRN,QZ	F&S	I	
CO1,CO2,CO3	Discuss research updates of Paradiguna.	CC	MK	KH	L,TP W,LS	T-EMI,P-VIV A,PRN,CR- RED	F&S	I	
Topic 4 4. Rasa (Lecture :7 hours, Non lecture: 4 hours)									
CO1,CO2,CO3	Define Rasa and interpret Meaning of “Rasa” in various contexts.	CK	MK	K	L&G D	P-VIVA,PUZ, M-CHT,INT	F&S	I	
CO1,CO2	Classify and compare shadarasa in relative correlation with taste of chemical constituents.	CK	MK	K	L&PP T,ED	P-VIVA,PUZ, M-CHT,INT	F&S	I	

CO1,CO2	Define Anurasa.	CK	MK	K	L,LS	T-EMI,P-VIVA,INT	F&S	I	
CO1,CO2	Interpret relevance of Anurasa in clinical practice.	CAP	MK	KH	L&G D,FC	T-EMI,P-VIVA,PUZ,INT	F&S	I	
CO1,CO2	Describe with justification Rasa sevanakrama as Aushadha.	CC	MK	KH	L&PP T,DIS	T-EMI,T- EW, P-VIVA,PRN, M-CHT,M- POS	F&S	I	
Topic 5 5. Vipaka (Lecture :3 hours, Non lecture: 1 hours)									
CO1	Define Vipaka.	CK	MK	K	L	P-VIVA	F&S	I	
CO1	State Vipaka lakshana.	CK	MK	K	L&G D	P-VIVA,INT	F&S	I	
CO1	Describe the action of Trividha Vipaka on Dosha, Dhatu and Mala with examples.	CC	MK	KH	L&PP T,TP W,FC	T- EW,P-VIVA,PRN	F&S	I	
CO1,CO3	Interpret clinical application of Vipaka.	CAP	MK	KH	L&PP T,DIS ,SDL, LS	P-VIVA,QZ	F&S	I	
CO2	State Research updates of Vipaka.	CK	MK	K	L&PP T	P-VIVA,CR- RED	F&S	I	
CO1	Explain Vipakopalabdh (Determination of Vipaka) and Taratamya (Degree of variation).	CC	MK	KH	L&PP T,FC	P-VIVA,PUZ, M-CHT	F&S	I	
Topic 6 6. Virya (Lecture :2 hours, Non lecture: 2 hours)									
CO1	Define the lakshanas of Virya.	CK	MK	K	L,SD	P-VIVA,INT	F&S	I	

CO1,CO2	Discuss research updates of Prabhava.	CC	MK	KH	L&G D,SD L	T- EW,P-VIV A,CR-RED	F&S	I	
Topic 8 8. Interrelation of Rasa-Guna-Virya-Vipaka-Prabhava (Lecture :1 hours, Non lecture: 2 hours)									
CO1	Interpret the interrelation of Rasa-Guna-Virya-Vipaka-Prabhava with respect to their strength- Pharmacodynamics.	CAP	MK	KH	L&PP T,TP W,PL	P-VIVA,CL- PR	F&S	I	
Topic 9 9. Karma (Lecture :9 hours, Non lecture: 5 hours)									
CO1	Define Karma.	CK	MK	K	L	T-OBT,P- VIVA,QZ	F&S	I	
CO1	Discuss Karma lakshana.	CC	MK	KH	L&G D	T-OBT,P- VIVA,QZ	F&S	I	
CO1,CO2,CO 4,CO5	Explain Deepan karma in relation with Appetizers.	CC	MK	KH	L&G D,BL	T- EW,T-OBT ,P-VIVA,QZ	F&S	I	
CO1,CO2,CO 3,CO4	Critically analyze Deepan dravyas with its gunas and application in various clinical conditions.	CAP	MK	KH	L&PP T,PB L,GB L	T-OBT,P- VIVA,QZ ,DEB	F&S	I	
CO1,CO2,CO 3,CO4	Explain Pachan karma in relation with Digestives.	CC	MK	KH	L&PP T	T-OBT,P- VIVA,QZ	F&S	I	
CO1,CO2,CO 3,CO4	Critically analyze Pachan dravyas with gunas and discuss its application in various clinical conditions.	CAN	MK	KH	L&PP T,PB L,PE R	T-EMI,T- OBT,P-VIVA	F&S	I	
CO1,CO2,CO	Explain Samshodhan karma in relation with contemporary	CC	MK	KH	L VC	T-OBT,P-	F&S	I	

4	pharmacological action.					,PL	VIVA,QZ			
CO1,CO3,CO4	Discuss Samshohan karma and its clinical application.	CC	MK	KH		PBL,RP,PER	T-EW,T-CS,T-OBT,P-VIVA	F&S	I	
CO1,CO2,CO3,CO4	Explain Samshamana karma in relation with contemporary pharmacological action.	CC	MK	KH		L&GD,BL	T-EMI,T-OBT,P-VIVA	F&S	I	
CO1,CO2,CO3,CO4	Describe Samshamana karma and its clinical application.	CC	MK	KH		L&PP T,CB L,PL	T-EMI,T-CS,P-VIVA,QZ	F&S	I	
CO1,CO2,CO3	Explain Anuloman karma in relation with Carminative.	CC	MK	KH		L_VC ,DIS	T-EMI,T-OBT ,P-VIVA,DEB	F&S	I	
CO1,CO3	Describe Anuloman karma and its clinical application.	CC	MK	KH		L&PP T,PB L,LS	T-OBT,P-VIVA,QZ	F&S	I	
CO1,CO2,CO3,CO4	Explain Sransana karma in relation with contemporary pharmacological action.	CC	MK	KH		L&PP T	T-EMI,T-OBT ,P-VIVA,DEB	F&S	I	
CO1,CO3	Discuss Sransan karma and its clinical application.	CC	MK	KH		L&PP T,PB L,LS	T-OBT,P-VIVA,QZ ,DEB	F&S	I	
CO1,CO2,CO3,CO4	Explain Bhedana karma in relation with strong laxative action	CC	MK	KH		L_VC ,FC	T-EMI,T-OBT ,P-VIVA,DEB	F&S	I	
CO1,CO3	Discuss Bhedan karma and its clinical application.	CC	MK	KH		L&PP T,DIS	T-OBT,P-VIVA,QZ	F&S	I	
CO1,CO2,CO3,CO4	Explain Rechana karma in relation with Purgative.	CC	MK	KH		L&PP T	T-EMI,T-OBT,P-VIVA	F&S	I	

CO1,CO3	Discuss Rechana karma and its clinical application.	CC	MK	KH	L&PP T,DIS	T-OBT,P- VIVA,QZ	F&S	I	
CO1,CO3	Discuss Chhedana karma and its clinical application.	CC	MK	KH	L&PP T,CB L,LS	T-EMI,T- OBT,P-VIVA	F&S	I	
CO1,CO2,CO 3,CO4	Explain Lekhana karma in relation with contemporary pharmacological action.	CC	MK	KH	L&G D,BL	T-EMI,T- OBT,P-VIVA	F&S	I	
CO1,CO3	Discuss Lekhana karma and its clinical application.	CC	MK	KH	L&PP T,DIS ,PL	T-OBT,P- VIVA,QZ ,DEB	F&S	I	
CO1,CO2,CO 3,CO4	Explain Grahi karma in relation with contemporary pharmacological action bowel binding.	CC	MK	KH	L VC	T-EMI,T- OBT,P-VIVA	F&S	I	
CO1,CO3	Discuss of Grahi karma and its clinical application.	CC	MK	KH	L&G D,TB L	T-OBT,P- VIVA,QZ	F&S	I	
CO1,CO2,CO 3,CO4	Explain Stambhana karma in relation with contemporary pharmacological action.	CC	MK	KH	L&PP T,FC	T-EMI,T-OBT ,P-VIVA,DEB	F&S	I	
CO1,CO3	Discuss Stambhana karma and its clinical application.	CC	MK	KH	L&G D,PB L	T-EMI,T-OBT ,P-VIVA,DEB	F&S	I	
CO1	Discuss Madakari karma with examples.	CC	MK	KH	L,DIS	T-OBT,P- VIVA,QZ	F&S	I	
CO1,CO3	Discuss Pramathi karma and its clinical application.	CC	MK	KH	L&PP T,DIS	T-EMI,T- OBT,P-VIVA	F&S	I	
CO1	Discuss Abhishyandi karma with examples.	CK	MK	K	L,DIS	T-EMI,T-OBT	F&S	I	

	therapeutic action of individual drugs.																			
Topic 11 11. Principles of General Pharmacology (Lecture :15 hours, Non lecture: 1 hours)																				
CO4,CO9	Define Pharmacology and discuss Principles of general Pharmacology.	CK	MK	K	L&PP T	PRN,QZ	F&S	I												
CO4,CO9	Discuss drug definition, drug dosage forms and route of drug administration.	CC	MK	KH	L_V	T- EW,P- VIVA,QZ	F&S	I												
CO4,CO9	Explain pharmacokinetics (ADME) drug and pharmacodynamics.	CC	MK	KH	L_V ,DIS	P-VIVA,PRN	F&S	I												
CO4,CO9	Discuss Drug dose, Principles of drug action, Mechanism of drug action & Bioavailability	CC	MK	KH	L_V ,PER	P-VIVA,QZ	F&S	I												
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on CNS with examples of Anaesthetics, Sedative-Hypnotic, Antiepileptic, Antiparkinsonian, Antidepressants, Antianxiety and Opioid Analgesics Drugs	CC	MK	KH	L&PP T,PB L,PrB L,FC	T- EW,T-OBT ,P-VIVA,QZ ,C-INT	F&S	III												
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Peripheral (somatic) Nervous System with examples of Skeletal Muscle Relaxants and Local Anaesthetics Drugs.	CC	MK	KH	L&PP T,PB L,FC	T- EW,T-OBT ,P-VIVA,QZ	F&S	III												
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Autacoids and Related of Nonsteroidal Antiinflammatory (NSAIDs), Antipyretic and Analgesics Drug	CC	MK	KH	L&PP T,PB L,FC	T- EW,T-OBT ,P-VIVA,PRN	F&S	III												
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Respiratory Disorders -	CC	MK	KH	L&PP T,PB	T-OBT,P- VIVA,PRN	F&S	III												

	Bronchodilators, Aerosols/ Inhalants, Expectorants and Antitussives Drugs					L,BL					
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Cardiovascular Drugs as Antihypertensive and Antianginal Drugs	CC	MK			KH	L_VC _FC	T-OBT,P- VIVA,PRN	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Kidney as Diuretics	CC	MK			KH	L&PP T,TP W,BL	T-OBT,P- VIVA,QZ	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting as Haematinics, Coagulants, Anticoagulants and Hypolipidaemic Drugs	CC	MK			KH	L&PP T,PB L,BL	T- EW,T- OBT,P-VIVA	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Gastrointestinal tract as Antacid, Carminatives, Digestants, Antiemetics, Laxatives, Antidiarrhoeal and Hepatoprotective Drugs	CC	MK			KH	L&PP T,L& GD,L _VC	T- EW,T-OBT ,P-VIVA,QZ	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting as Antibiotics and Antitubercular.	CC	MK			KH	L&PP T,FC, GBL	T- EW,P- VIVA,QZ	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting as Antifungal, Antiviral, Antimalarial and Anthelmintic Drugs	CC	MK			KH	L&PP T,PB L,GB L	T- EW,T-OBT ,P-VIVA,QZ	F&S	III	
CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on Hormones and Related Drugs as Thyroid Hormone, Thyroid Inhibitors, Insulins, Oral Antidiabetic, Hormonal Contraceptives, Uterine Stimulants and Uterine Relaxants Drugs.	CC	MK			KH	L&PP T,BL, PER	T- EW,T-OBT ,P-VIVA,PRN	F&S	III	

CO9	Define, Describe mode of action & Discuss types with examples of following drugs acting on as Antiseptics , Disinfectants, Vaccines, Vitamins, Water imbalance and IV	CC	MK	KH	L&PP T,BL, PER	T- EW,T- OBT,P-VIVA	F&S	III	
Topic 12 12. Mishraka Gana (Lecture :6 hours, Non lecture: 2 hours)									
CO1,CO3	Explain Brihatpanchamoola composition.	CC	MK	KH	L&PP T,ED U	T- EW,P-VIV A,P-POS,QZ ,M-CHT	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Brihatpanchamoola.	CC	MK	KH	L&PP T,DIS ,TBL	T- EW,P-VIV A,P-POS,QZ ,M-CHT	F&S	III	
CO1,CO3	Explain Laghupanchamoola composition.	CC	MK	KH	L&G D	T- EW,P-VIV A,P-POS,QZ ,M-CHT	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Laghupanchamoola.	CC	MK	KH	L&PP T,DIS ,PL	T- EW,P-VIV A,P-POS,QZ	F&S	III	
CO1,CO3	Explain Vallipanchamoola composition.	CC	MK	KH	L&PP T	T- EW,P-VIV A,P-POS,QZ ,M-CHT	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Vallipanchamoola.	CC	MK	KH	L&G D,PE R	T- EW,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Kantakapanchamoola composition.	CC	MK	KH	L&PP T,BS,	T- EW,P- VIVA,QZ	F&S	III	

CO1,CO3	Describe Guna karma and combined therapeutic effect of Trimada.	CC	MK	KH	L&PP T,DIS ,TUT	T- EW,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Chaturusana composition.	CC	MK	KH	L&PP T	T-OBT,P- VIVA,P-POS	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Chaturusana.	CC	MK	KH	L&PP T,FC	P-VIVA,P- POS,QZ	F&S	III	
CO1,CO3	Explain Panchakola composition.	CC	MK	KH	L&PP T,L_ VC	P-VIVA,QZ	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Panchakola.	CC	MK	KH	L&PP T,DIS	T- EW,T-CS,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Shadusana composition.	CC	MK	KH	L&PP T,PL	T-OBT,P- VIVA,P-POS	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Shadusana	CC	MK	KH	L&PP T,FC	T- EW,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Chaturbeeja composition.	CC	MK	KH	L&PP T,L_ VC	T-OBT,P-VIV A,P-EXAM	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Chaturbeeja	CC	MK	KH	L&PP T,PE R	T- EW,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Trijataka composition.	CC	MK	KH	L_ VC ,DIS, SDL	T-OBT,P-VIV A,P-EXAM,P- POS	F&S	III	

CO1,CO3	Describe Guna karma and combined therapeutic effect of Trijataka	CC	MK	KH	L&PP T,PL	T- EW,P- VIVA,P-POS	F&S	III	
CO1,CO3	Explain Chaturajataka composition.	CC	MK	KH	L_VC ,PER	T-OBT,P- VIVA,QZ	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Chaturajataka.	CC	MK	KH	L&PP T,DIS	T- EW,P- VIVA,P-POS	F&S	III	
CO1,CO3	Explain Panchatikta composition.	CC	MK	KH	L&PP T,FC	T-OBT,P-VIV A,P-REC,QZ	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Panchatikta.	CC	MK	KH	L&PP T,PL	T- EW,P- VIVA,P-POS	F&S	III	
CO1,CO3	Explain Chaturbhadra composition.	CK	MK	K	L,DIS	T-OBT,P- VIVA,QZ	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Chaturbhadra.	CC	MK	KH	L&PP T,FC	T-EMI,P- VIVA,QZ	F&S	III	
CO1,CO3	Explain Trikarshika composition.	CC	MK	KH	L,DIS	T-OBT,P-VIV A,P-EXAM	F&S	III	
CO1,CO3	Describe Guna karma and combined therapeutic effect of Trikarshika.	CC	MK	KH	L&PP T,DIS ,PL	T-OBT,P- VIVA,QZ	F&S	III	
Topic 13 13. Nomenclature of dravya as per Nighantu, Vedic taxonomy and Botany (Lecture :0 hours, Non lecture: 2 hours)									
CO1	Describe the Nomenclature of dravya as per Raj Nighantu.	CC	MK	KH	L&PP T,FC, RP	P-VIVA,PRN, P-ID,QZ	F&S	III	
CO1	Explain the Nomenclature of dravya as per Vedic taxonomy.	CC	MK	KH	L&PP T	P-VIVA,P- ID,QZ	F&S	III	

CO1	Discuss the Nomenclature of dravya as per botany.	CC	MK	KH	L&PP T,BL, PER	P-VIVA,P- ID,QZ	F&S	III	
Topic 14 14. Prashasta Bhesaja, Bhesaja Pariksha and drug evaluation method with correlation as per Pharmacognosy (Lecture :1 hours, Non lecture: 2 hours)									
CO1,CO5	Describe Prashasta Bhesaja.	CC	MK	KH	L&G D,PL	P-VIVA,QZ	F&S	III	
CO1,CO5	Explain Bhesaja Pariksha of Charaka samhita vimana sthana.	CC	MK	KH	L&G D,SD L,LS	P-VIVA,QZ ,M-CHT	F&S	III	
CO1,CO5	Discuss drug evaluation method in correlation with Pharmacognosy.	CC	MK	KH	L&PP T,DIS	P-VIVA,QZ	F&S	III	
Topic 15 15. Dravyasangrahana and Drug collection methods as per GFCP (Good Field collection practices) (Lecture :1 hours, Non lecture: 0 hours)									
CO1,CO2	Discuss Dravyasangrahana and Drug collection methods as per GFCP (Good Field collection practices).	CC	DK	KH	L&PP T,PE R	P-VIVA,PUZ, CHK	F&S	III	
CO1,CO2	Discuss Drug collection methods as per GFCP (Good Field collection practices).	CC	DK	KH	L&PP T,DIS ,PrBL	P-VIVA,PUZ, CHK	F&S	III	
Topic 16 16. GCP (Good cultivation practices), seed bank, conservation of medicinal plants, knowledge about RET (Rear, Endangered & Threatened) medicinal plants. (Lecture :1 hours, Non lecture: 0 hours)									
CO2	Discuss good cultivation practices, seed bank, conservation of medicinal plants, knowledge about RET (Rear, Endangered & Threatened) medicinal plants.	CC	DK	KH	L&PP T,L_ VC,F C	P-VIVA,M- POS	F&S	III	

Topic 17 17. Abhava Pratinidhi Dravya (substitutes) (Lecture :1 hours, Non lecture: 1 hours)							
CO1,CO7	Discuss the concept of Abhava pratinidhi dravya (substitutes) as per Bhavaprakasha.	CC	MK	KH	L&G D,BS, EDU	P-VIVA,QZ	F&S III
Topic 18 18. Classifications and techniques of aqueous and alcoholic extracts (Lecture :0 hours, Non lecture: 2 hours)							
CO2,CO6	Appraise the techniques of aqueous and alcoholic extracts of medicinal plants.	CC	DK	KH	L_V _D	P- VIVA,DOPS	F&S III
Topic 19 19. Adverse drug reaction and Pharmacovigilance with recent updates (Lecture :1 hours, Non lecture: 2 hours)							
CO2,CO8	Explain adverse drug reaction and pharmacovigilance in ayurveda with recent updates.	CC	MK	KH	L&PP T,L& GD,L S	P-VIVA,PRN, QZ ,M-CHT	F&S III V-RS
Topic 20 20. NMPB (National Medicinal Plant Board), CCRAS (Central Council of Research in Ayurveda Sciences), API (Ayurvedic Pharmacopeia of India), GCTM (Global Centre for Traditional Medicine), PCIMH (Pharmacopeia Commission of Indian Medicine and Homeopathy) (Lecture :1 hours, Non lecture: 0 hours)							
CO2	State NMPB (National Medicinal Plant Board), CCRAS (Central Council of Research in Ayurveda Sciences), API (Ayurvedic Pharmacopeia of India), GCTM (Global Centre for Traditional Medicine) and PCIMH (Pharmacopeia Commission of Indian Medicine and Homeopathy).	CK	NK	K	SDL	P-VIVA,QZ ,INT	F&S III
Topic 21 21. Vrikshayurveda and Ethno-medicine (Lecture :1 hours, Non lecture: 1 hours)							
CO2	Explain concept of Vrikshayurveda and Ethno medicine.	CC	NK	KH	L&G D,ML ,PER	P-VIVA,CR- RED	F&S III
Topic 22 22. Network pharmacology and Bioinformatics (Lecture :1 hours, Non lecture: 1 hours)							

CO2	Describe Network Pharmacology and Bioinformatics.	CC	NK	KH	L&G D,SD L,LS	P-VIVA,CR- RED	F&S	III	
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Paper 2 Applied Dravyaguna									
A3	B3	C3	D3	E3	F3	G3	H3	I3	J3
Course outcome	Learning Objective (At the end of the session, the students should be able to)	Doma in/sub	Must to know / desirable to know / Nice to know	Level Does/ Shows how/ Knows how/ Know	T-L method	Assessment (Refer abbreviations)	Formative/ summative	Term	Integration
Topic 1 1. Bheshajavacharaniya (Criteria’s to be considered for selection of drugs in vyadhis) (Lecture :1 hours, Non lecture: 6 hours)									
CO5	Interpret the selection of appropriate drugs in different vyadhis as per criteria’s mentioned in Bheshajavacharaniya (as per As. Sa. Su 23)	CAP	MK	KH	L&PP T,CB L,PrB L,SD L	T- EW,P- VIVA,QZ	F&S	II	H-RN ,H- SW
Topic 2 2.1 Dravya (Drug) Nama-Guna-Karma Jnana (Lecture :45 hours, Non lecture: 10 hours)									
CO3	Specify useful parts with its Rasapanchaka of following drugs.	CK	MK	K	L&G D,CB L,FC	T-OBT,P- VIVA	F&S	II	
CO4	Describe karma,agryakarma and dosha karma of following drugs.	CC	MK	KH	L&PP	T-EMI,T-OBT	F&S	II	

	various Srotas and vyadhiavastha (Stage).					T,DIS ,CBL	VIVA,QZ			
CO5	Indicate the Kalpana(dosage form), Matra (Dose), Anupana (Vehicle), Marga (Route), Sevana kala (Time of administration), Kalavadhi (Duration) and Pathya-pathya) of following drugs.	CAP	MK	KH		L&PP T,DIS ,ML	T-EMI,P- VIVA	F&S	III	V-RS, H-SW
CO2	Enlist active phyto-constituents & important formulations. Discuss research updates of following drugs.	CK	MK	K		L&G D,BS	T-EMI,T-OBT ,P-VIVA,QZ	F&S	III	
CO8	Enlist botanical name & family. Explain main Synonyms as per Bruhatryees and Bhavaprakasha. Vernacular name (Hindi, English and local name) of following drugs.	CK	MK	K		L&G D,FC, ML	T-OBT,P- VIVA,QZ	F&S	III	
CO8	Demonstrate external morphology-habit, root, leaf, stem, flower, inflorescence, fruit, seed and officinal useful parts of following drugs.	CC	MK	KH		L_VC ,ML	T-OBT,P- VIVA,QZ	F&S	III	
CO7	Describe varieties, grhyahrahyatwa, adulterants, substitute and toxic effects of following drugs wherever applicable.	CC	MK	KH		L&PP T,DIS ,EDU	T-EMI,T-OBT ,P-VIVA,PUZ	F&S	III	H-AT